GOA UNIVERSITY

Department of Computer Science and Technology

MINI PROJECT REPORT

EXPENSE TRACKER

A Personal Expense Management and Budget Planning Application

Submitted in partial fulfilment of the requirements for the mini project of the

MASTER OF COMPUTER APPLICATIONS (MCA)

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DECLARATION

We, Obaid Shaikh (Roll No. 2631) and Sanish Kalangutkar (Roll No. 2644), Omkar Bordekar (Roll No. 2633) hereby declare that the mini project entitled “EXPENSE TRACKER” is a record of the work carried out by us.

This project has been prepared as part of our academic requirements. The information, ideas, references, and resources obtained from other sources have been acknowledged wherever applicable.

We declare that the contents of this report accurately represent the work completed by our team, and that any proposed or incomplete features are identified accordingly.

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We would also like to thank our classmates and everyone who provided useful feedback and suggestions during the planning and development of the application.

This project has provided us with an opportunity to apply our academic knowledge to a practical problem and gain experience in software development, teamwork, and project documentation.

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Project Summary

The Expense Tracker is a personal finance management application designed to help users monitor their daily expenses, manage monthly budgets, understand spending patterns, and plan their savings more effectively. The project aims to go beyond basic transaction recording by providing features that help users reflect on their spending behaviour and make informed financial decisions.

The proposed application includes features such as an impulse purchase detector, hidden small-expense finder, goal impact calculator, streak-based engagement, expense categorisation, festival expense planning, emergency savings tracking, expense stories, and subscription plans.

The application is intended to help users identify the cumulative effect of small purchases, understand how spending decisions affect savings goals, and prepare for irregular expenses such as festivals and emergencies. Its behaviour-focused approach aims to make personal budgeting more understandable and engaging.

The project involves requirement analysis, system design, feature planning, application development, and testing. The final implementation and results are documented in this report, along with the system diagrams, screenshots, testing outcomes, limitations, and possible future enhancements.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Managing personal finances is an important part of everyday life. However, tracking expenses can be difficult due to frequent transactions, digital payments, and small purchases that often go unnoticed. These expenses can accumulate over time and affect monthly budgets and savings.

The Expense Tracker is a personal finance management application designed to help users record, organise, and understand their spending habits. It focuses not only on recording expenses but also on providing meaningful insights into spending behaviour and helping users make better financial decisions.

The project follows three main principles: behaviour-first, goal-linked, and India-aware. It aims to identify spending patterns, connect expenses with financial goals, and support planning for festivals and seasonal expenses.

1.2 Background of the Project

The increasing use of smartphones and digital payments has made transactions convenient, but it can also make spending difficult to monitor. Traditional methods, such as notebooks and spreadsheets, require regular manual effort and may not provide meaningful insights into spending habits.

The Expense Tracker aims to address this problem by combining expense recording with budget planning and financial awareness. Features such as small-expense detection, goal impact calculation, and festival expense planning are intended to help users understand their spending and prepare for future financial needs.

1.3 Motivation

The main motivation behind this project is to help users understand where their money goes and how their spending decisions affect their savings. Frequent small purchases and unplanned expenses may seem insignificant but can influence financial goals over time.

The project also aims to encourage regular expense tracking and improve financial planning. Its key motivations are:

1. To improve awareness of daily spending habits.
2. To identify the cumulative impact of small expenses.
3. To connect spending decisions with savings goals.
4. To encourage consistent expense tracking.
5. To support festival and emergency expense planning.

1.4 Project Overview

The Expense Tracker focuses on organising expenses and helping users understand their financial behaviour. Its main proposed features include:

1. Impulse Purchase Detector: Helps users identify potentially unplanned purchases.
2. Hidden Small-Expense Finder: Highlights repeated small expenses and their cumulative impact.
3. Goal Impact Calculator: Shows how spending may affect progress towards financial goals.
4. Streak-Based Engagement: Encourages users to track expenses regularly.
5. Festival Expense Planner: Helps users plan for seasonal and festival-related spending.
6. Emergency Savings Tracker: Helps users monitor progress towards an emergency fund.
7. Narrative Expense Story: Presents spending patterns in an easy-to-understand format.
8. Subscription Plans: Represents the proposed paid or tiered access model for application services.

These features aim to make expense management more understandable and encourage informed financial decisions. The final report will distinguish between features implemented in the application and those that remain proposed.

1.5 Target Audience

The Expense Tracker is designed for individuals who want to manage their expenses and improve their financial awareness. Its primary target audience includes:

- Students: To monitor daily spending and manage limited funds.
- Young Earners: To balance personal expenses, budgets, and savings.
- Early-Career Professionals: To understand spending patterns and plan financial goals.
- Goal-Driven Savers: To track progress towards goals such as travel, gadgets, or emergency savings.
- Festival and Seasonal Spenders: To prepare for additional expenses during special occasions.
- Smartphone Users: To manage personal finances conveniently through a digital application.

The application aims to provide these users with a structured way to understand their spending habits, plan expenses, and work towards their financial goals.

Chapter Summary

This chapter introduced the Expense Tracker project, its background, motivation, proposed features, and target audience. The project focuses on combining expense management with spending awareness, financial goal planning, and preparation for seasonal expenses. The next chapter presents the problem statement, objectives, scope, benefits, and limitations of the project.

CHAPTER 2

PROBLEM STATEMENT, OBJECTIVES AND SCOPE

2.1 Problem Statement

Managing daily expenses can be challenging due to frequent transactions, unplanned purchases, and small expenses that are often overlooked. Although users may record their spending, they may not clearly understand how their spending habits affect their monthly budget and savings goals.

Existing expense management methods may focus mainly on recording transactions and displaying totals. Users may also need assistance in identifying repeated small expenses, understanding the impact of purchases on financial goals, and planning for additional expenses during festivals and special occasions.

Therefore, the project aims to develop an **Expense Tracker application** that helps users organise their expenses, understand spending patterns, and plan their finances through behavioural insights, goal-oriented tracking, and expense planning features.

2.2 Objectives of the Project

The main objective of the Expense Tracker project is to improve personal expense management and financial awareness.

The specific objectives are:

1. To provide a structured method for recording and categorising expenses.
2. To help users identify frequent small expenses and potentially impulsive purchases.
3. To show how spending decisions may affect financial goals.
4. To encourage regular expense tracking through streak-based engagement.
5. To support planning for festival and seasonal expenses.
6. To help users monitor progress towards emergency savings goals.
7. To present spending information in a simple and understandable format.
8. To explore subscription plans as a possible model for providing application services.

2.3 Scope of the Project

The scope of the Expense Tracker project covers personal expense management, spending analysis, and basic financial planning. The application is intended to help users organise

their financial information and understand their spending behaviour.

The proposed scope includes the following areas:

- **Expense Management:** Recording and categorising daily expenses.
- **Spending Analysis:** Identifying repeated small expenses and potentially unplanned purchases.
- **Goal Planning:** Understanding the relationship between spending and savings targets.
- **Budget Planning:** Supporting the organisation of spending limits and planned expenses.
- **Festival Planning:** Preparing for expenses associated with festivals and seasonal occasions.
- **Emergency Savings:** Monitoring progress towards an emergency fund.
- **User Engagement:** Encouraging regular interaction through streak-based features.
- **Financial Insights:** Presenting spending patterns through readable summaries or expense stories.
- **Subscription Model:** Considering different access plans for application features.

The project primarily focuses on personal finance awareness rather than advanced banking or investment management. Features requiring automatic bank integration, payment

processing, or external financial data are outside the scope unless specifically implemented.

2.4 Benefits of the Project

The Expense Tracker is intended to provide the following benefits:

1. **Improved Financial Awareness:** Helps users understand how their money is spent.
2. **Better Expense Organisation:** Groups expenses into meaningful categories.
3. **Identification of Small Expenses:** Highlights the combined effect of frequent low-value purchases.
4. **Goal-Oriented Spending:** Helps users understand the relationship between expenditure and savings targets.
5. **Improved Financial Planning:** Supports preparation for festivals, seasonal expenses, and emergencies.
6. **Consistent Tracking:** Streak-based engagement aims to encourage regular expense recording.
7. **Easy Understanding:** Presents financial information in a structured and readable manner.

These benefits depend on the features available in the final implemented version of the application.

2.5 Limitations of the Project

Although the Expense Tracker aims to simplify personal expense management, certain limitations may apply.

1. **Manual Data Entry:** If expenses are entered manually, the accuracy of the analysis depends on the information provided by users.
2. **Limited Financial Integration:** Automatic synchronisation with bank accounts or payment applications may not be available.
3. **Dependence on User Participation:** Regular tracking requires users to record expenses and review the information.
4. **Accuracy of Insights:** Spending insights depend on the completeness and correctness of recorded transactions.
5. **Limited Financial Scope:** The project focuses on personal expenses and savings awareness rather than professional accounting, investment advice, or comprehensive banking services.
6. **Feature Availability:** Some proposed features may require further development before they can be used in the application.

The limitations described above should be reviewed and adjusted according to the actual implementation.

Chapter Summary

This chapter described the problem addressed by the Expense Tracker, its main objectives, scope, benefits, and limitations. The project aims to improve personal expense

management by combining expense organisation with spending awareness and goal-based planning. The next chapter examines existing expense management applications and compares their features with the proposed Expense Tracker.

CHAPTER 3 – EXISTING SYSTEM AND COMPETITIVE ANALYSIS

3.1 Introduction

Personal finance applications help users record expenses, manage budgets, and monitor their financial activities. However, users may still struggle to identify spending habits, track small expenses, and plan for savings goals.

The **Expense Tracker** aims to address these challenges through expense management, behavioural insights, and goal-oriented financial planning.

3.2 Existing System

Traditional expense tracking involves notebooks, spreadsheets, or mobile applications. Manual methods require regular effort, while spreadsheets often need users to analyse the data themselves. Existing applications provide digital expense tracking and budgeting tools, but their features differ according to their primary purpose.

3.3 Overview of Existing Applications

- **ET Money:** Focuses on personal finance and investment-related services.
- **INDmoney:** Provides investment tracking and wealth management features.

- **YNAB:** Emphasises structured budgeting and planned spending.
- **PocketGuard:** Helps users monitor spending and available funds.

These applications provide useful financial tools. The proposed Expense Tracker focuses more on spending behaviour, small-expense awareness, and festival-related planning.

3.4 Comparative Analysis

Feature	ET Money	INDmoney	YNAB	PocketGuard	Expense Tracker
Impulse Purchase Detector	No	No	No	No	Yes
Small-Expense Finder	No	Partial	No	No	Yes
Goal Impact Calculator	Partial	Yes	Partial	No	Yes
Streak-Based Engagement	No	No	No	No	Yes
Festival Expense Planner	No	No	No	No	Yes
Emergency Savings Tracker	Partial	Yes	Partial	Yes	Yes
Narrative Expense Story	No	No	No	No	Yes

3.5 Limitations and Opportunities

The project identifies the following opportunities for improvement in expense management:

1. **Spending Awareness:** Identifying potentially impulsive purchases.
2. **Small-Expense Analysis:** Showing the cumulative impact of repeated small purchases.
3. **Goal-Based Planning:** Understanding how expenses affect savings goals.
4. **Seasonal Budgeting:** Planning expenses for festivals and special occasions.
5. **User Engagement:** Encouraging regular expense tracking.
6. **Savings Monitoring:** Tracking progress towards an emergency fund.

3.6 Proposed System

The Expense Tracker is designed to help users organise expenses and understand spending patterns. Its proposed features include an impulse purchase detector, hidden small-expense finder, goal impact calculator, streak-based engagement, custom categories, festival expense planner, emergency savings tracker, narrative expense story, and subscription plans.

These features aim to make expense tracking more informative and encourage better financial awareness.

3.7 Advantages of the Proposed System

- Improves awareness of spending habits.
- Organises expenses into categories.
- Highlights the cumulative impact of small purchases.
- Connects spending with savings goals.
- Supports festival and emergency expense planning.
- Encourages consistent expense tracking.

3.8 Summary

This chapter reviewed existing expense management methods and compared four personal finance applications. The analysis identified opportunities to improve behavioural spending awareness, goal-based planning, and seasonal budgeting. The proposed Expense Tracker aims to address these needs through additional expense analysis and planning features.

CHAPTER 4 – REQUIREMENTS AND FEASIBILITY ANALYSIS

4.1 Introduction

Requirements analysis identifies the functions and resources needed to develop the Expense Tracker application. This chapter describes the functional and non-functional requirements, system requirements, and feasibility of the proposed project.

4.2 Functional Requirements

The proposed system should support the following functions:

1. Expense Management: Record and organise expenses.
2. Spending Analysis: Identify spending patterns and small expenses.
3. Goal Planning: Help users understand the impact of expenses on savings goals.
4. Budget Planning: Support planning for regular and festival-related expenses.
5. Savings Tracking: Monitor progress towards emergency savings goals.
6. User Engagement: Encourage regular expense tracking through streaks.
7. Expense Summaries: Present spending information in an understandable format.

8. Subscription Plans: Support different access plans if implemented.

4.3 Non-Functional Requirements

- Usability: The application should be easy to understand and navigate.
- Performance: Expense records and summaries should load efficiently.
- Reliability: Expense information should be stored and retrieved accurately.
- Security: Personal financial information should be protected.
- Maintainability: The application should be easy to update and improve.
- Compatibility: The application should work on its intended target platform.

4.4 Hardware and Software Requirements

The exact requirements depend on the technologies used to develop the application.

Category	Requirement
Hardware	Computer or laptop for development

Memory	Sufficient RAM to run the development tools
Storage	Sufficient space for source code, dependencies, and project data
Software	Operating system compatible with the development environment
Development Tools	Project-specific IDE or code editor
Testing	Target device, emulator, or browser, depending on the application

4.5 Feasibility Analysis

4.5.1 Technical Feasibility

The project is technically feasible if the selected development tools, database, and platform support expense recording, data storage, and the required analysis features.

4.5.2 Economic Feasibility

The project can be developed using available development tools and resources. Actual costs depend on the selected technologies, hosting requirements, and any paid services used.

4.5.3 Operational Feasibility

The application is intended to provide a simple way for users to record expenses, monitor spending, and plan savings. A clear interface can help users understand and use its features.

4.5.4 Time Feasibility

The project can be developed in stages, beginning with essential expense management and followed by analysis and planning features. The final scope depends on the available development time.

4.6 Summary

This chapter outlined the functional and non-functional requirements, general system requirements, and feasibility of the Expense Tracker project. The analysis provides a foundation for system design and implementation.

CHAPTER 5 – BRAINSTORMING AND IDEA SELECTION

5.1 Introduction

Brainstorming is the process of generating and evaluating ideas to identify suitable solutions to a problem. For the Expense Tracker project, brainstorming focused on improving expense management, increasing spending awareness, and helping users plan their financial goals.

5.2 Brainstorming Ideas

The following ideas were considered during the planning of the project:

1. Impulse Purchase Detector: Identify potentially unplanned purchases.
2. Hidden Small-Expense Finder: Highlight the cumulative impact of small expenses.
3. Goal Impact Calculator: Show how spending affects savings goals.
4. Streak-Based Engagement: Encourage regular expense tracking.
5. Custom Spending Categories: Organise expenses into relevant categories.
6. Festival Expense Planner: Help users plan seasonal and festival expenses.

7. Emergency Savings Tracker: Monitor progress towards an emergency fund.
8. Narrative Expense Story: Present spending patterns through simple summaries.
9. Subscription Plans: Explore different levels of application access.

5.3 Idea Evaluation and Selection

The ideas were considered based on their usefulness, relevance to the target audience, feasibility, and potential to improve the expense-tracking experience.

Idea	Purpose
Impulse Purchase Detector	Improve spending awareness
Hidden Small-Expense Finder	Identify cumulative small expenses
Goal Impact Calculator	Connect spending with financial goals
Streak-Based Engagement	Encourage regular tracking
Custom Spending Categories	Organise expense records
Festival Expense Planner	Support seasonal budgeting

Emergency Savings Tracker	Monitor savings progress
Narrative Expense Story	Simplify spending analysis
Subscription Plans	Explore potential monetisation

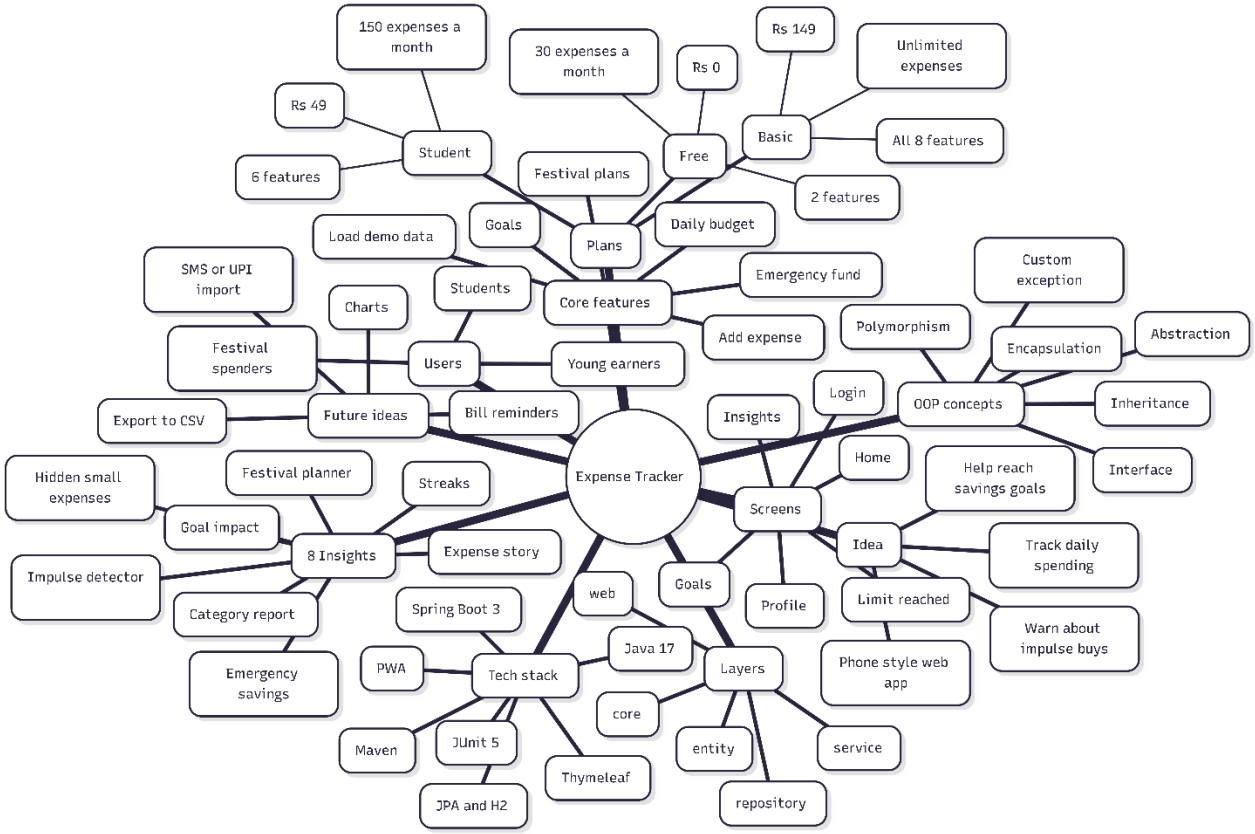
The selected ideas were intended to make the application more useful than a basic expense-recording tool. Their implementation depends on the final project scope and available development time.

5.4 Rationale for the Selected Idea

The Expense Tracker was selected because it combines everyday expense management with spending awareness and financial planning. Its focus on small expenses, savings goals, and festival-related budgeting makes it relevant to students and young earners who want to understand and manage their spending.

5.5 Summary

This chapter described the brainstorming process and the main ideas considered for the Expense Tracker. The selected concepts aim to improve expense organisation, spending awareness, and goal-based financial planning. The next chapter presents the system design and modelling.


Expense Tracker
Plans
Free
Rs 0
30 expenses a month
Basic
Rs 149
Unlimited expenses
All 8 features
2 features
Student
150 expenses a month
Rs 49
6 features
Festival plans
Goals
Daily budget
Core features
Students
Load demo data
Emergency fund
Add expense
Users
Young earners
Charts
SMS or UPI import
Festival spenders
Future ideas
Bill reminders
Export to CSV
Screens
Insights
Login
Home
Goals
Profile
Idea
Track daily spending
Limit reached
Warn about impulse buys
Phone style web app
Help reach savings goals
OOP concepts
Custom exception
Polymorphism
Encapsulation
Abstraction
Inheritance
Interface
Layers
core
entity
service
repository
Tech stack
web
Java 17
Spring Boot 3
PWA
Maven
JUnit 5
Thymeleaf
JPA and H2
8 Insights
Streaks
Festival planner
Hidden small expenses
Goal impact
Expense story
Impulse detector
Category report
Emergency savings

CHAPTER 6 - SYSTEM DESIGN AND MODELLING

6.1 Introduction

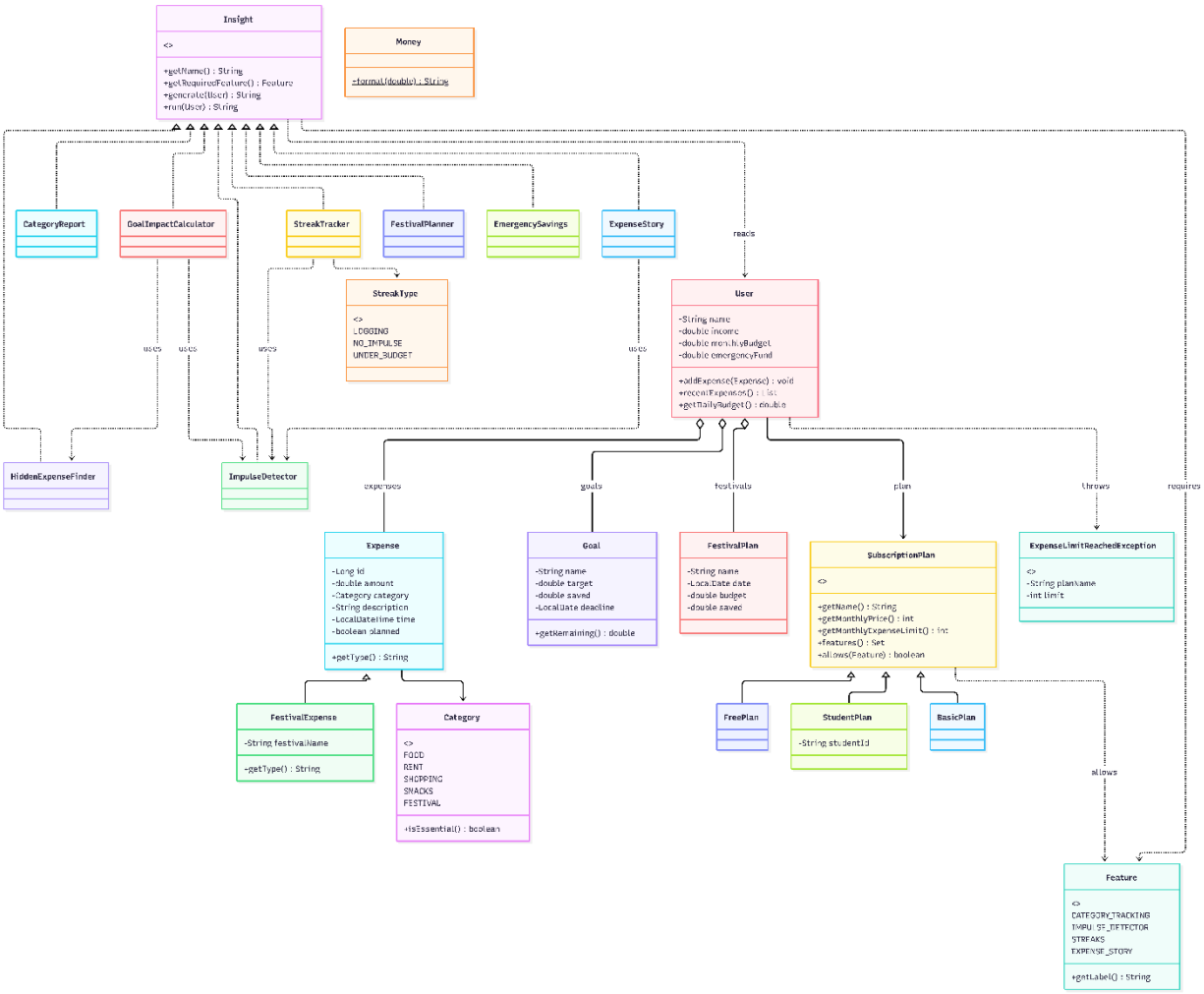
System design and modelling describe the structure, behaviour and data organisation of the Expense Tracker application. The diagrams in this chapter illustrate the application's object-oriented design, interaction between software components, user activities and database structure.

The system uses Java 17 and Spring Boot 3, with a core domain layer for business logic and separate layers for web handling and database operations. UML diagrams are used to represent the relationships between classes and explain how the application processes user requests.

6.2 Core Domain Class Diagram

The Core Domain Class Diagram represents the main classes, interfaces, enumerations and relationships used in the Expense Tracker's business logic.

The User class manages user-related information and maintains relationships with expenses, goals, festival plans and subscription plans. The Expense class represents an expense record, while FestivalExpense extends Expense to represent festival-related expenses.

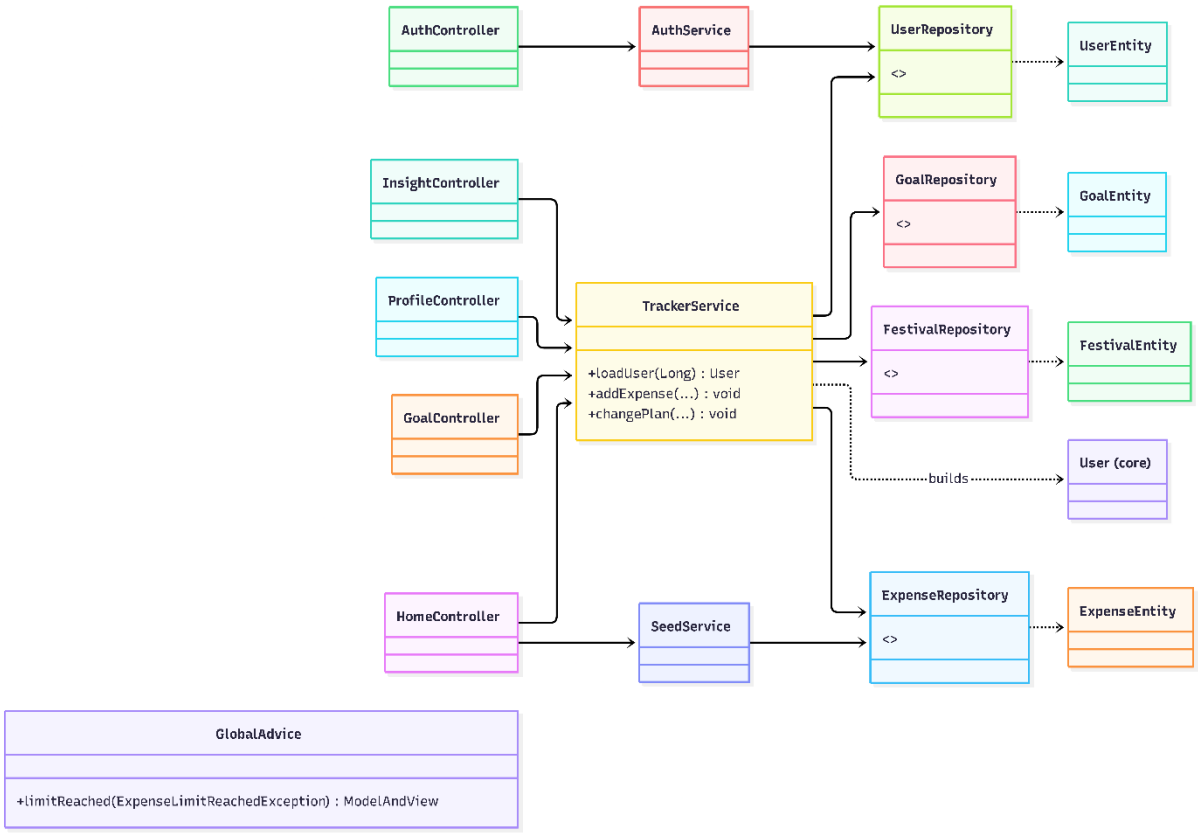


The abstract Subscription Plan class defines common subscription-related behaviour. The Free Plan, Student Plan and BasicPlan classes inherit from it. The Insight interface defines a common structure for the application's insight implementations, including category reporting, impulse detection, hidden expense identification, goal impact calculation, streak tracking, festival planning, emergency savings and expense story generation.

The diagram demonstrates object-oriented concepts such as encapsulation, inheritance, abstraction, interfaces and

polymorphism. It also shows how domain classes interact with one another to implement the application's business rules.

6.3 Application Layers Class Diagram



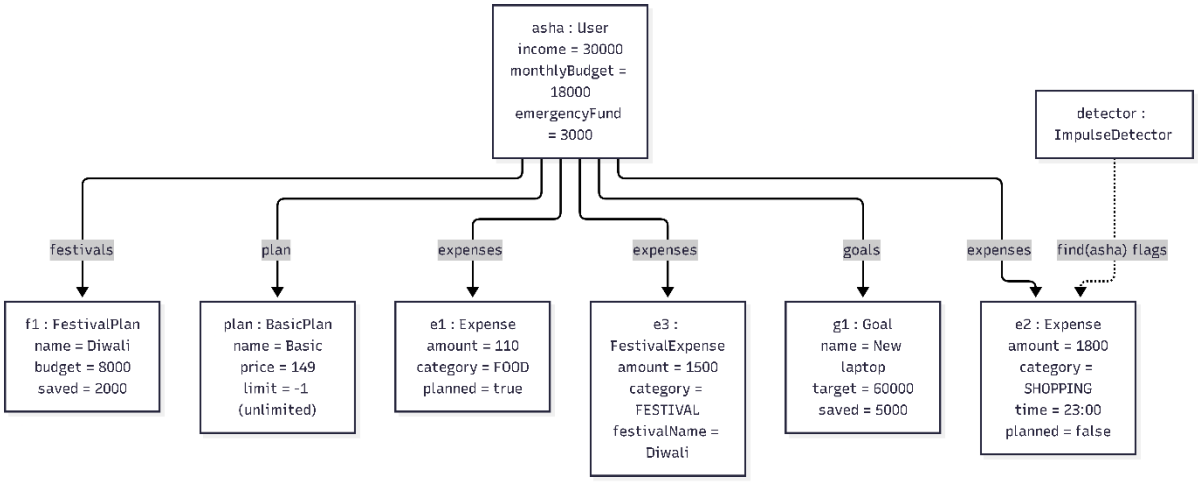
The Application Layers Class Diagram illustrates the organisation of the web, service, repository, entity and core domain components.

The web layer contains controllers responsible for handling user requests. The service layer coordinates application operations and connects the controllers with the data-access components. The repository layer provides database access

through Spring Data interfaces, while the entity layer represents persistent database records.

The service layer also constructs core domain objects to execute business logic. This separation helps organise the application and keeps the core domain classes independent of web framework code.

6.4 Object Diagram



The Object Diagram presents an example snapshot of objects in memory when demonstration data is loaded for a user subscribed to the Basic plan.

It illustrates a user object linked to a subscription plan, expense objects, a goal object and a festival plan object. It also shows an insight object interacting with expense information.

The diagram provides an example of how instances of the domain classes are associated at runtime. Unlike a class diagram, which represents general class definitions and

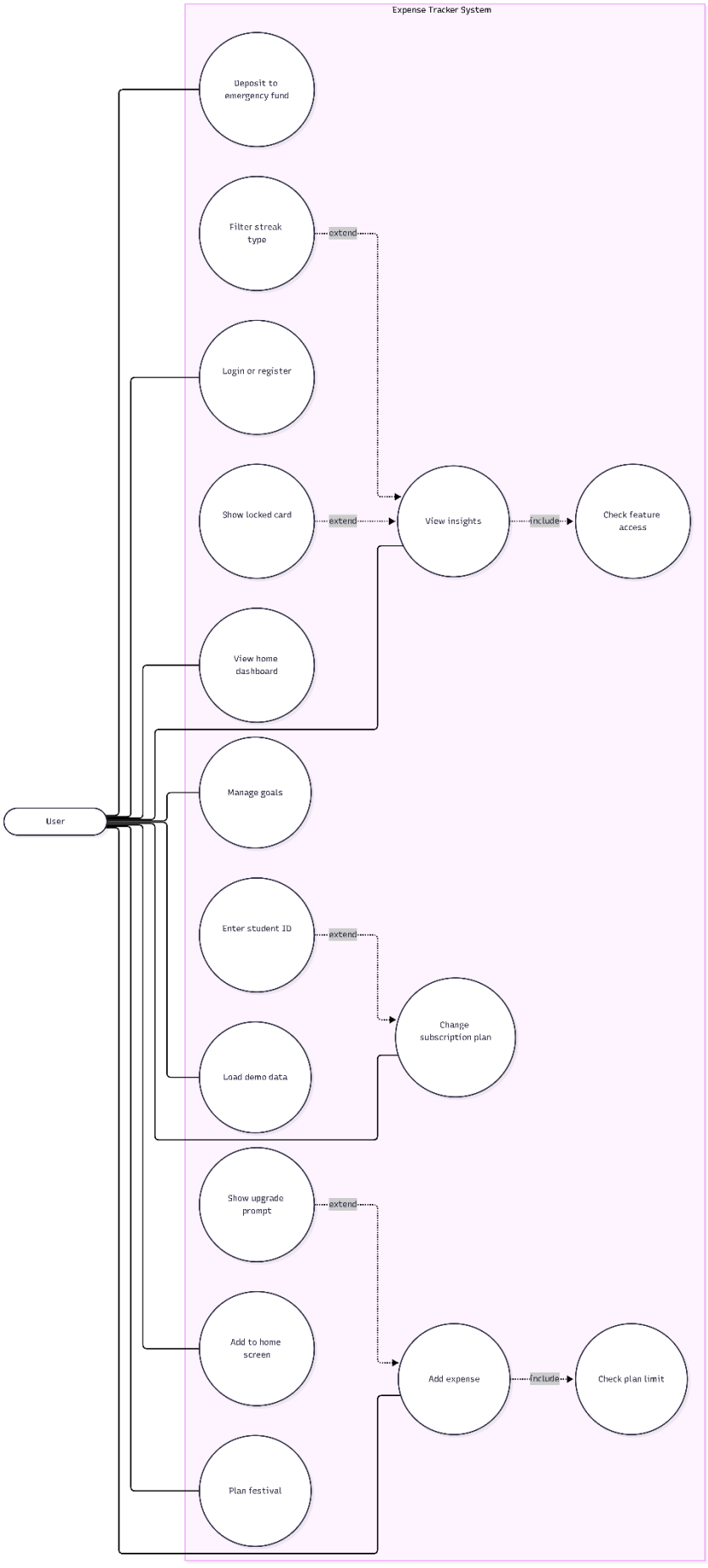
relationships, an object diagram represents specific instances and example attribute values.

6.5 Use Case Diagram

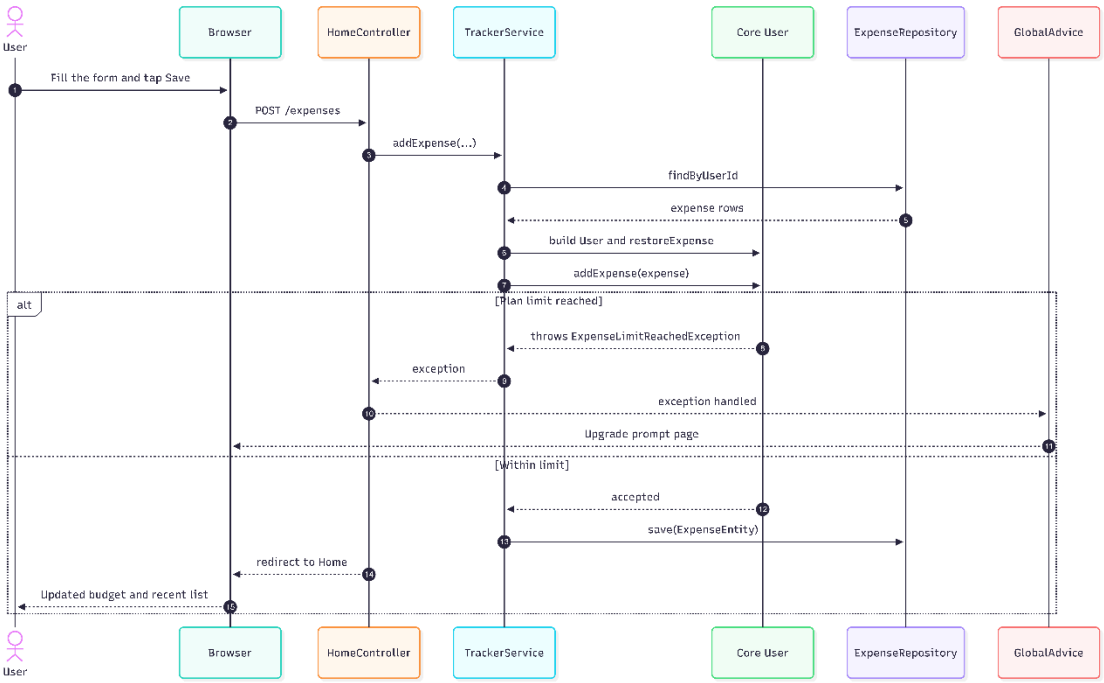
The Use Case Diagram represents the major functions available to a user of the Expense Tracker application.

The user can log in or register, add expenses, view the home dashboard, access insights, manage goals, plan festival expenses, deposit money into the emergency fund and change the subscription plan. The diagram also represents supporting interactions, including checking subscription limits, checking feature access, displaying upgrade prompts and loading demonstration data.

The diagram provides an overview of the interaction between the user and the system. It helps identify the application's main functional requirements and shows how subscription restrictions affect access to selected features.


Expense Tracker System
Deposit to emergency fund
Filter streak type
extend
Login or register
Show locked card
extend
View insights
include
Check feature access
View home dashboard
Manage goals
User
Enter student ID
extend
Change subscription plan
Load demo data
Show upgrade prompt
extend
Add to home screen
Add expense
include
Check plan limit
Plan festival

6.6 Sequence Diagram



The Sequence Diagram illustrates the sequence of interactions involved in adding an expense.

The user enters expense details in the browser and submits the form. The request is received by the HomeController, which communicates with TrackerService. The service retrieves the relevant expense records through the repository and constructs the core User object required for business-rule processing.

The core logic checks whether the expense can be accepted under the applicable subscription limit. If the limit is reached, an ExpenseLimitReachedException is raised and handled by the global exception-handling component, which displays an upgrade prompt. If the expense is accepted, it is saved

through the repository, and the user is redirected to the home page.

The diagram explains the communication between the user interface, controller, service, core domain and repository layers during expense entry.

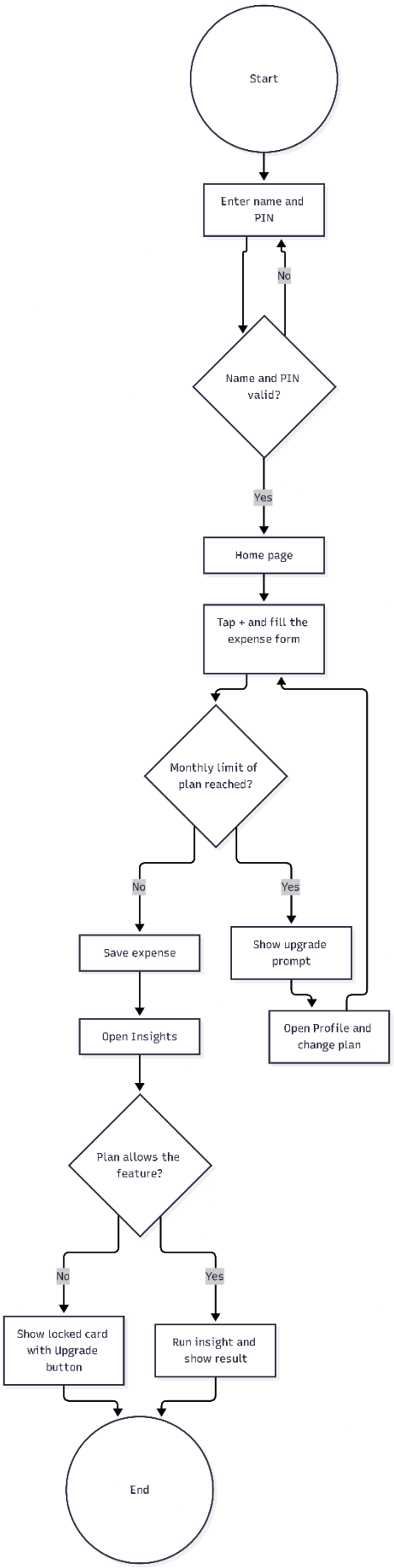
6.7 Activity Diagram

6.7.1 Activity Diagram: Using the Application

This activity diagram represents the main workflow from login to adding an expense and viewing insights.

The process begins when the user enters their login details. After successful access, the user opens the home page and enters expense information. The system checks the applicable subscription limit. If the limit has been reached, the application displays an upgrade prompt and allows the user to change their plan. Otherwise, the expense is saved.

The user can then open the insights section. The system checks whether the current subscription plan allows access to the selected feature. Depending on the result, the application displays either the insight or a locked feature card with an upgrade option.


Start
Enter name and PIN
No
Name and PIN valid?
Yes
Home page
Tap + and fill the expense form
Monthly limit of plan reached?
No
Yes
Save expense
Show upgrade prompt
Open Insights
Open Profile and change plan
Plan allows the feature?
No
Yes
Show locked card with Upgrade button
Run insight and show result
End

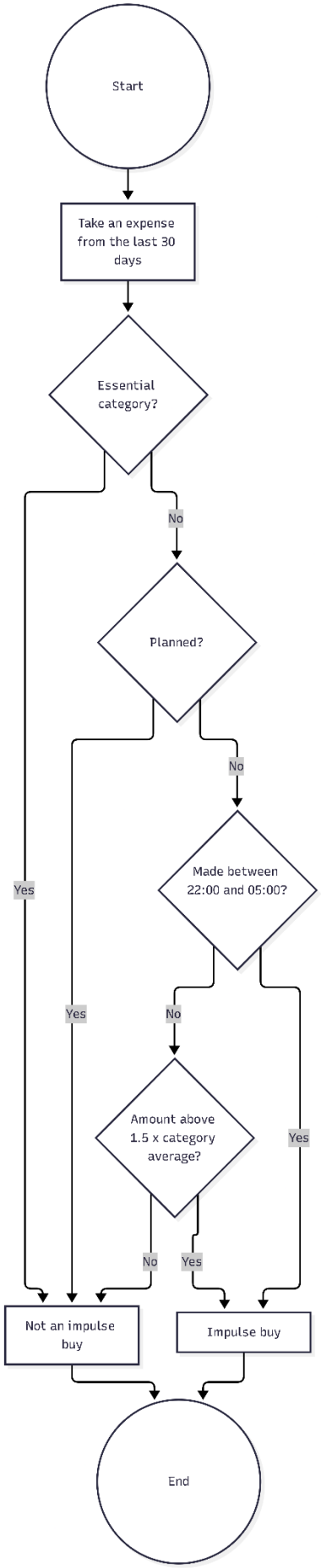
6.7.2 Activity Diagram: Impulse Purchase Detection

This activity diagram explains the decision-making process used by the impulse purchase detection feature.

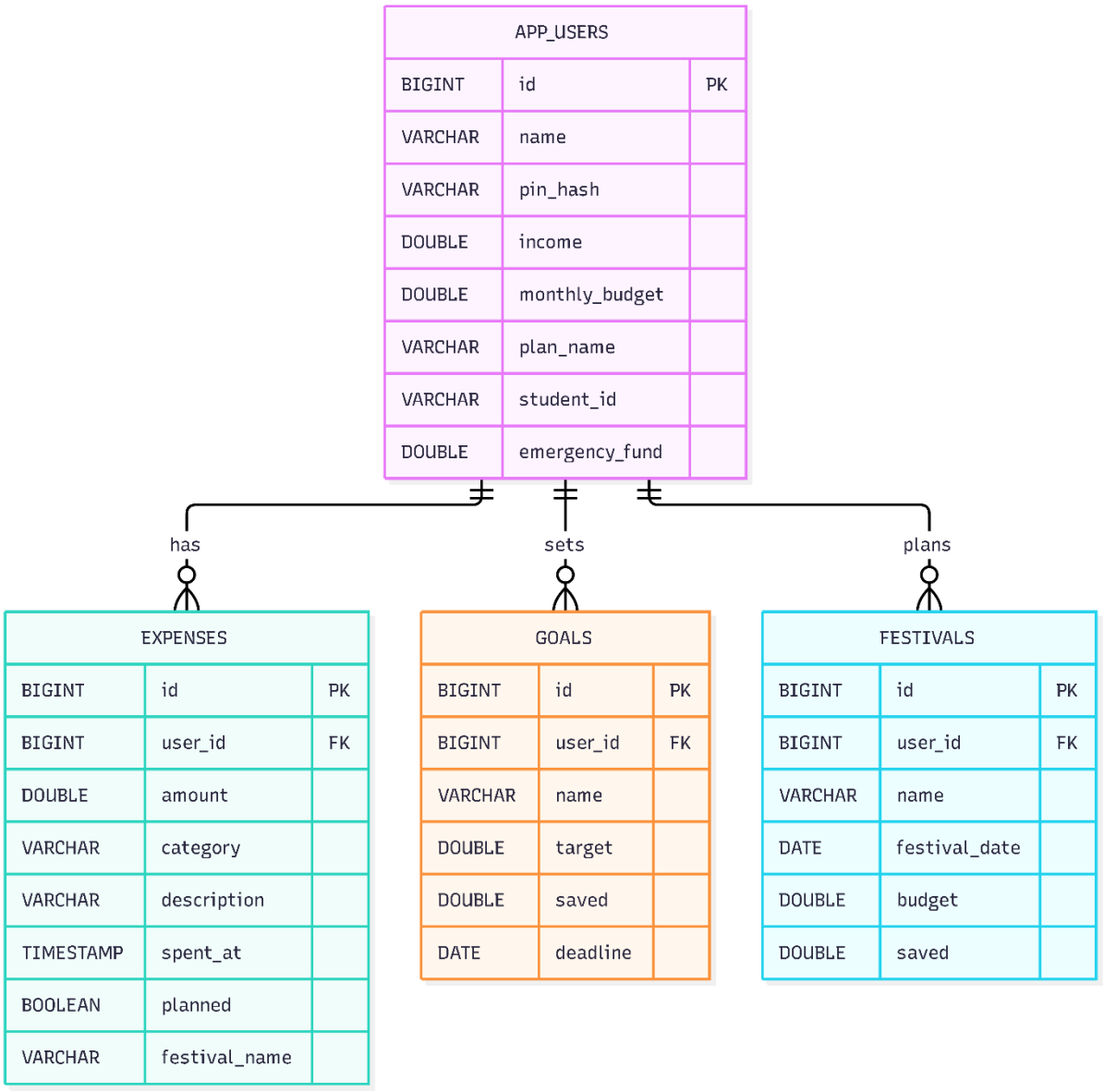
The process examines an expense and checks whether its category is essential, whether the expense was planned and whether it meets the application's impulse detection conditions.

According to the rules represented in the diagram, a non-essential and unplanned expense is classified as an impulse purchase if it was made between 10:00 PM and 5:00 AM or its amount exceeds 1.5 times the category average. Otherwise, it is not classified as an impulse purchase.

The diagram illustrates how conditional logic is used to classify expenses according to predefined rules.


Start
Take an expense from the last 30 days
Essential category?
Yes
No
Planned?
Yes
No
Made between 22:00 and 05:00?
Yes
No
Amount above 1.5 x category average?
Yes
No
Yes
Not an impulse buy
Impulse buy
End

6.8 Entity Relationship Diagram



The Entity Relationship Diagram represents the database tables and their logical relationships in the Expense Tracker application.

The diagram contains four tables:

- APP_USERS: Stores user information, login-related data, income, monthly budget, subscription plan and emergency fund information.
- EXPENSES: Stores expense amounts, categories, descriptions, timestamps, planned status and related festival information.
- GOALS: Stores savings goal details, including the goal name, target amount, saved amount and deadline.
- FESTIVALS: Stores festival planning information, including the festival name, date, budget and saved amount.

Each table has an identifier represented as its primary key. The ER diagram shows logical relationships between users and their expense, goal and festival records through the user_id column.

These relationships illustrate how records are associated with a particular user. The diagram's notes specify that the user_id links are represented as logical relationships rather than database foreign-key constraints.

6.10 Chapter Summary

This chapter presented the design and modelling of the Expense Tracker using the available UML diagrams and ER diagram. The core domain class diagram illustrated the object-oriented design, while the application layers class

diagram explained the organisation of the software components.

The object diagram showed example object relationships, and the use case diagram described the main user functions. The sequence and activity diagrams explained the expense-entry workflow and impulse detection logic. Finally, the ER diagram and database design section described the database tables and their logical relationships.

Together, these diagrams provide an overview of the application's structure, behaviour and data organisation.

CHAPTER 7 – FEATURES AND MODULES

7.1 Introduction

The Expense Tracker is designed to help users organise expenses, understand spending patterns, and plan their finances. This chapter describes the main features proposed for the application and their intended purposes.

7.2 Expense Management

This module allows users to record and organise their expenses. Categorising expenses helps users review their spending and understand where their money is being used.

7.3 Impulse Purchase Detector

This feature aims to identify potentially impulsive purchases based on available expense information. It is intended to encourage users to review their spending decisions and develop better financial awareness.

7.4 Hidden Small-Expense Finder

Small purchases may appear insignificant individually but can contribute considerably to total expenditure over time. This feature highlights repeated small expenses and their cumulative impact.

7.5 Goal Impact Calculator

The Goal Impact Calculator aims to show how spending decisions may affect progress towards savings goals. It helps users relate everyday expenses to their financial objectives.

7.6 Streak-Based Engagement

This feature encourages users to record expenses consistently. Tracking activity over consecutive days can help users develop a regular expense-monitoring habit.

7.7 Custom Spending Categories

This module allows expenses to be organised into categories such as food, transportation, shopping, and entertainment. Custom categories can make expense summaries more relevant to individual users.

7.8 Festival Expense Planner

The Festival Expense Planner is intended to help users prepare budgets for festivals and other seasonal occasions. It can help users plan expected expenses and monitor spending against their budgets.

7.9 Emergency Savings Tracker

This feature helps users monitor progress towards an emergency savings target. It provides a way to compare current savings with the intended target.

7.8 Narrative Expense Story

The Narrative Expense Story aims to present spending patterns through simple, readable summaries. This can make financial information easier to understand than viewing transaction figures alone.

7.9 Subscription Plans

The proposed subscription model consists of Three plans:

Plan	Price	Expenses/month	Features
Free	Rs. 0	30	Category report, Streaks
Student	Rs. 49	150	+ Impulse, Hidden expenses, Goal impact, Emergency savings (needs Student ID)
Basic	Rs. 149	Unlimited	All 8 (+ Festival planner, Expense story)

7.10 Summary

This chapter described the key modules and proposed features of the Expense Tracker. Together, these features aim to support expense organisation, spending awareness, savings planning, and consistent financial tracking.

8. Implementation and Technologies Used

The Expense Tracker is a mobile-style web application developed using Java 17 and Spring Boot 3. It follows a layered architecture that separates core business logic from web handling and database operations. Thymeleaf is used to render the user interface, while Spring Data JPA provides database access. The application demonstrates object-oriented programming through inheritance, abstraction, interfaces, encapsulation and polymorphism.

8.1 Technology Stack

Area	Technology
Programming Language	Java 17
Framework	Spring Boot 3
Build Tool	Apache Maven
Template Engine	Thymeleaf
Data Access	Spring Data JPA
Database	H2 file-based database
Frontend	HTML, CSS and JavaScript
Installable Web App	Web manifest and service worker
Testing Framework	JUnit 5

Development Tools	Visual Studio Code, Git and GitHub

8.2 Frontend Implementation

The frontend is implemented using Thymeleaf templates, HTML and CSS. The interface follows a mobile-style layout intended to resemble an expense-tracking application on a smartphone. JavaScript is used for selected interactive functions.

The application provides pages for accessing the home screen, viewing insights, adding expenses, managing goals and accessing profile-related options. The interface also displays subscription-related feature restrictions where applicable.

The project includes a web manifest and service worker to support installation-related functionality. Full Progressive Web App installation depends on the browser and connection requirements.

8.3 Backend and Database Implementation

The backend follows a layered architecture consisting of five main packages:

- core: Contains the domain classes, subscription plans, insight implementations and custom exception.
- entity: Contains Java Persistence API (JPA) entity classes representing database records.

- repository: Contains Spring Data repository interfaces for database access.
- service: Coordinates database operations and invokes the core business logic.
- web: Contains web controllers, login interception and global exception handling.

The application uses an H2 file-based database to store persistent application data. The project documentation identifies four database tables: APP_USERS, EXPENSES, GOALS and FESTIVALS.

Spring Data JPA and Hibernate are used for persistence. The service layer loads database records and constructs core domain objects before applying business rules. The application stores its database files in the local ./data directory.

8.4 User Authentication and Data Management

The project documentation describes a login process in which a user enters a name and a PIN containing four to six digits. A new name creates an account.

The application uses a login interceptor to protect requests that require an authenticated session. The web layer also includes exception handling through a controller advice component.

Users can access the expense-tracking features and load demonstration data through the application. The database contains user, expense, goal and festival records.

Verification note: Confirm the PIN hashing algorithm, session handling, logout behaviour and user-level data isolation from the source code before documenting their precise implementation details.

8.5 Expense Calculation Logic

The application provides eight insight implementations through a common Insight interface. These include category reporting, impulse detection, hidden expense identification, goal impact calculation, streak tracking, festival planning, emergency savings and expense story generation.

The insight classes encapsulate the rules used to analyse expense information and produce results. The application uses rule-based domain logic to provide spending-related information and support budgeting decisions.

The common interface allows different insight implementations to be handled consistently. The subscription plan determines whether a particular insight is available to the user.

Verification note: Include exact calculation formulas, date ranges, thresholds and badge rules only after checking the corresponding source-code methods.

8.6 Budget and Savings Calculations

The project includes domain classes for users and goals, together with insight implementations related to goal impact, festival planning and emergency savings.

These components support the application's budgeting and savings-related functionality. The core domain layer contains the relevant business rules, while the service layer connects those rules with the stored application data.

The goal and savings calculations should be documented using the actual formulas implemented in the source code. Any formula included in the final report must match the application's behaviour and should not be described as a machine-learning prediction.

8.7 Module Integration

The application integrates its components through a layered design. Requests are handled by the web layer, which communicates with the service layer. The service layer accesses the repositories and constructs core domain objects to execute business rules.

The Insight interface is implemented by eight insight classes. Its default run() method includes subscription-based feature access checks. The application also defines a custom checked exception, ExpenseLimitReachedException, for expense-limit handling. The web layer uses global exception handling to present an upgrade prompt when the relevant exception is handled.

This separation improves code organisation and allows the core business logic to be tested independently of the web interface.

Chapter Summary

The Expense Tracker combines Java, Spring Boot, Thymeleaf, Spring Data JPA and H2 to implement an object-oriented expense-tracking web application. Its layered architecture separates domain logic, persistence and web handling. The common insight interface and subscription plan hierarchy demonstrate important object-oriented programming concepts.

CHAPTER 9 – TEST CASES

9.1 Introduction

Testing is performed to verify that the Expense Tracker behaves according to its requirements. It helps identify errors in expense processing, subscription restrictions, insight calculations, data management and interaction between application components.

The project uses JUnit 5 for testing the core domain layer. The repository documentation specifies 25 unit tests and provides Maven as the test execution tool. Additional manual testing is required to evaluate the web interface and complete application workflows.

9.2 Testing Approach

The following testing activities are relevant to the project:

1. Unit Testing

Unit testing checks individual core classes and their business rules independently. The project documentation identifies 25 JUnit 5 tests for the core domain layer.

2. Functional Testing

Functional testing checks whether the application's features behave as expected when users perform operations such as logging in, adding expenses, viewing insights and managing goals.

3. Database Testing

Database testing verifies whether application records are stored and retrieved correctly using the H2 file-based database.

4. Integration Testing

Integration testing checks communication between the web controllers, service layer, repositories, database and core domain classes.

The functional, database and integration tests listed above must be performed before their results are reported as completed.

9.3 Test Cases

The following table provides a test plan. Actual results and status must be filled in after executing each test.

Test ID	Test Scenario	Expected Result	Actual Result / Status
TC01	Run the JUnit 5 test suite using mvn test.	Test execution completes and the report displays the actual pass/fail counts.	Pending execution
TC02	Log in using a valid name and a PIN of the accepted length.	The application allows access to the appropriate page.	Pending manual test
TC03	Attempt to access a protected page without a valid session.	The login flow is enforced according to the application configuration.	Pending manual test

TC04	Add an expense within the subscription plan's limit.	The expense is accepted and saved if the operation is valid.	Pending manual test
TC05	Attempt to exceed the applicable expense limit.	The limit-handling logic responds as implemented, including the upgrade prompt where applicable.	Pending manual test
TC06	View an insight allowed by the current subscription plan.	The corresponding insight is processed and its result is displayed.	Pending manual test
TC07	Attempt to access an insight restricted by the current plan.	Access is handled according to the subscription rules.	Pending manual test
TC08	Create or update a savings goal.	Valid goal information is processed and persisted as implemented.	Pending manual test
TC09	Load demonstration data.	The application loads the available sample data for demonstrating its features.	Pending manual test
TC10	Restart the application and check previously saved records.	Records persist as expected in the H2 file-based database.	Pending manual test

9.4 Test Results

The repository documentation confirms that the project includes 25 JUnit 5 tests for the core domain layer. However, the test execution output must be examined to determine how many tests pass or fail.

The results of manual functional testing, database testing and integration testing must also be recorded after the tests have been executed. No pass rate or successful outcome should be claimed without supporting test evidence.

After execution, the test summary should report:

- Total automated tests executed.
- Number of tests passed.
- Number of tests failed or skipped.
- Important defects identified and their status.
- Manual test results supported by observed application behaviour.

9.5 Issues and Limitations

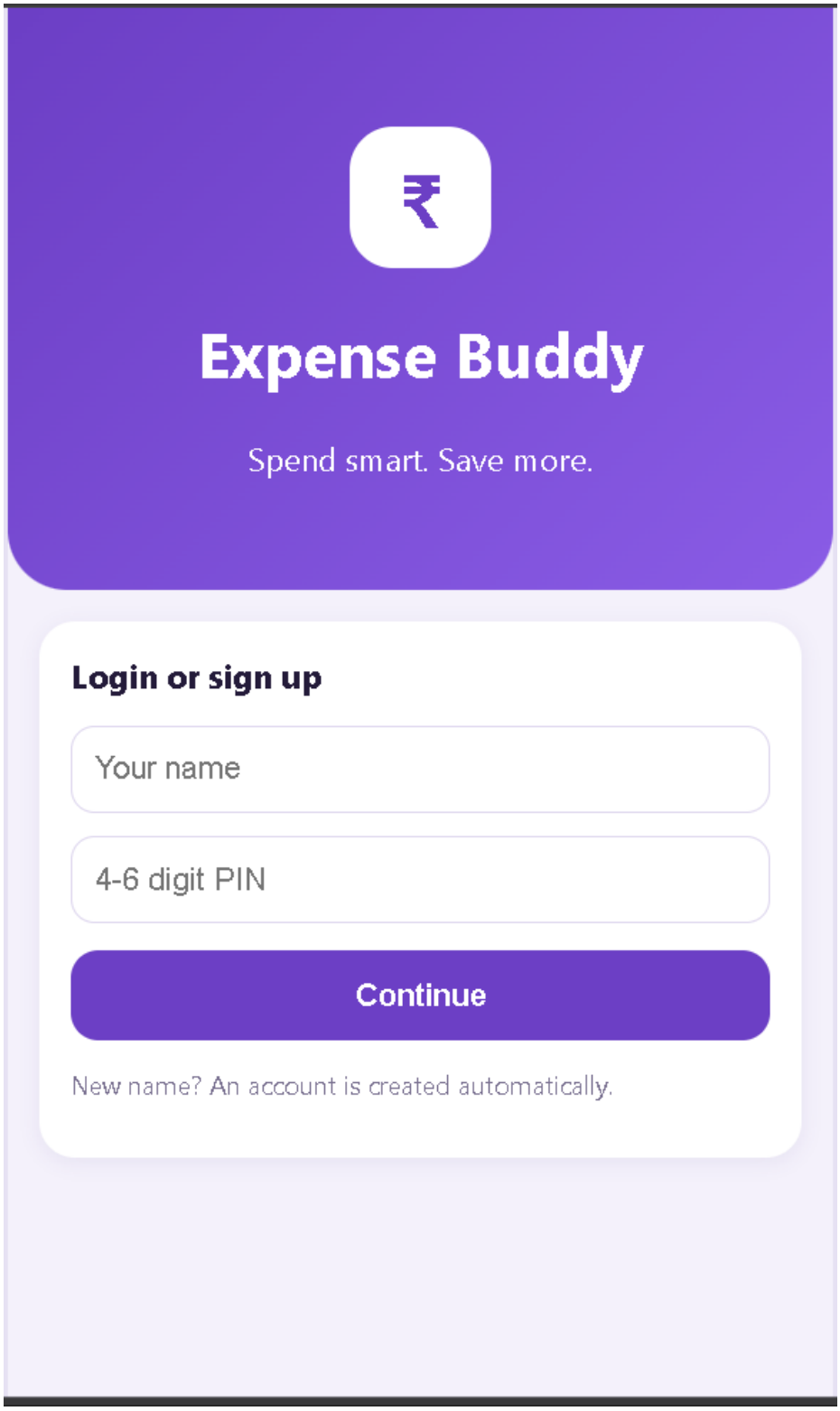
Any defects encountered during testing should be documented with their symptoms, cause where known, and corrective action. Features that have not been tested should be identified as unverified rather than assumed to work correctly.

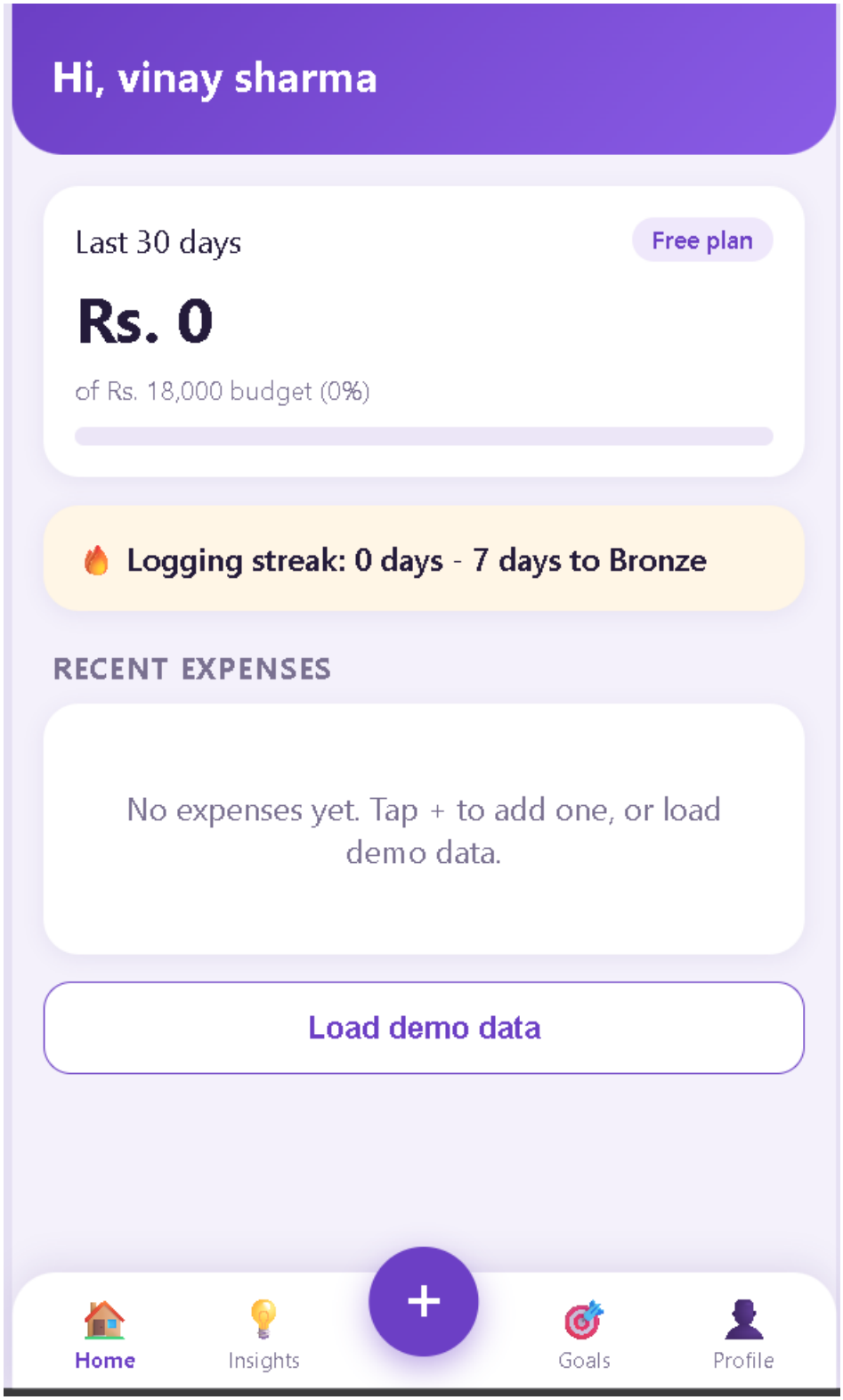
Testing of the core domain layer does not, by itself, establish that the complete web application is free of errors. Browser behaviour, database persistence, session handling and integration between application layers require appropriate verification.

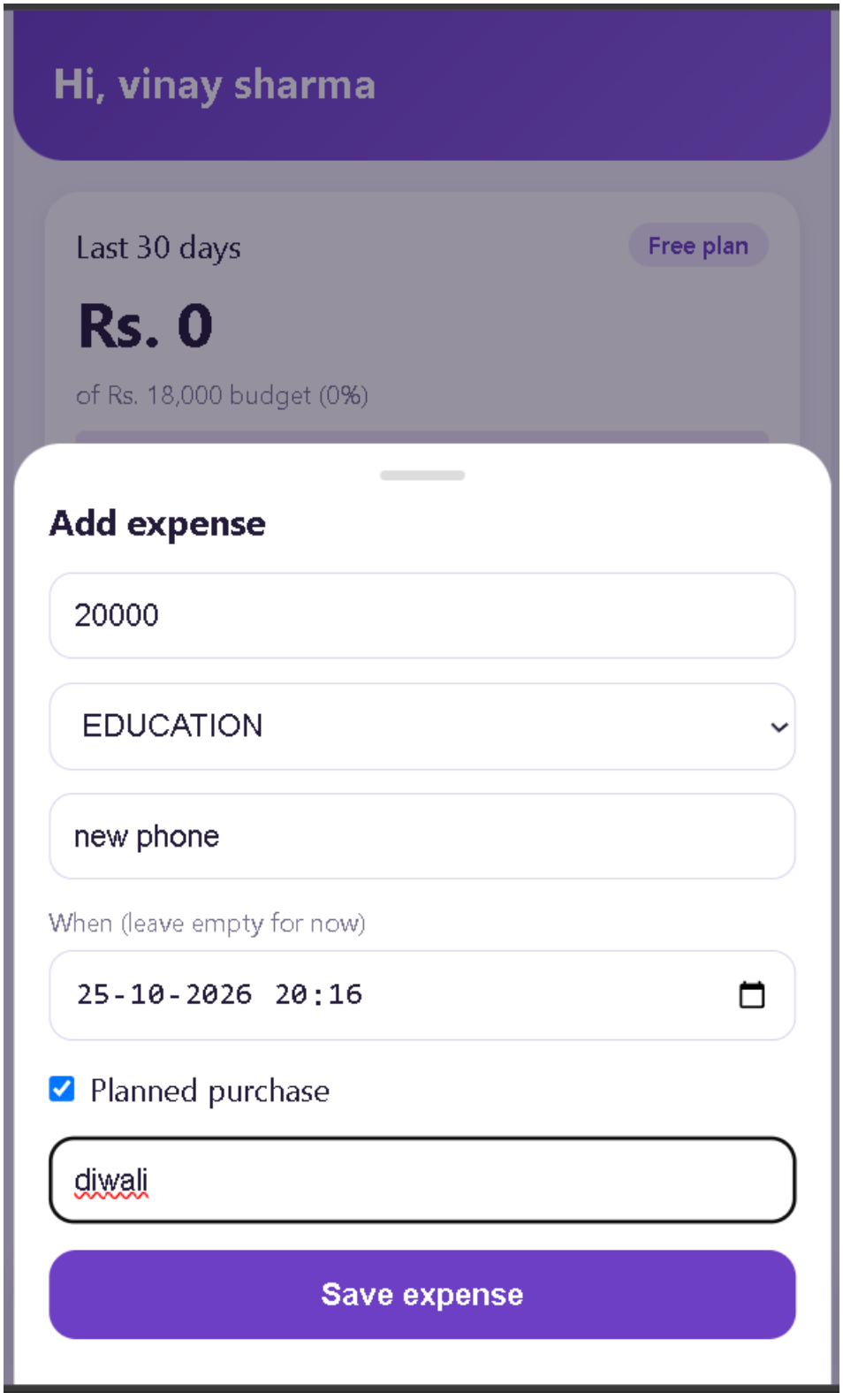
9.6 Chapter Summary

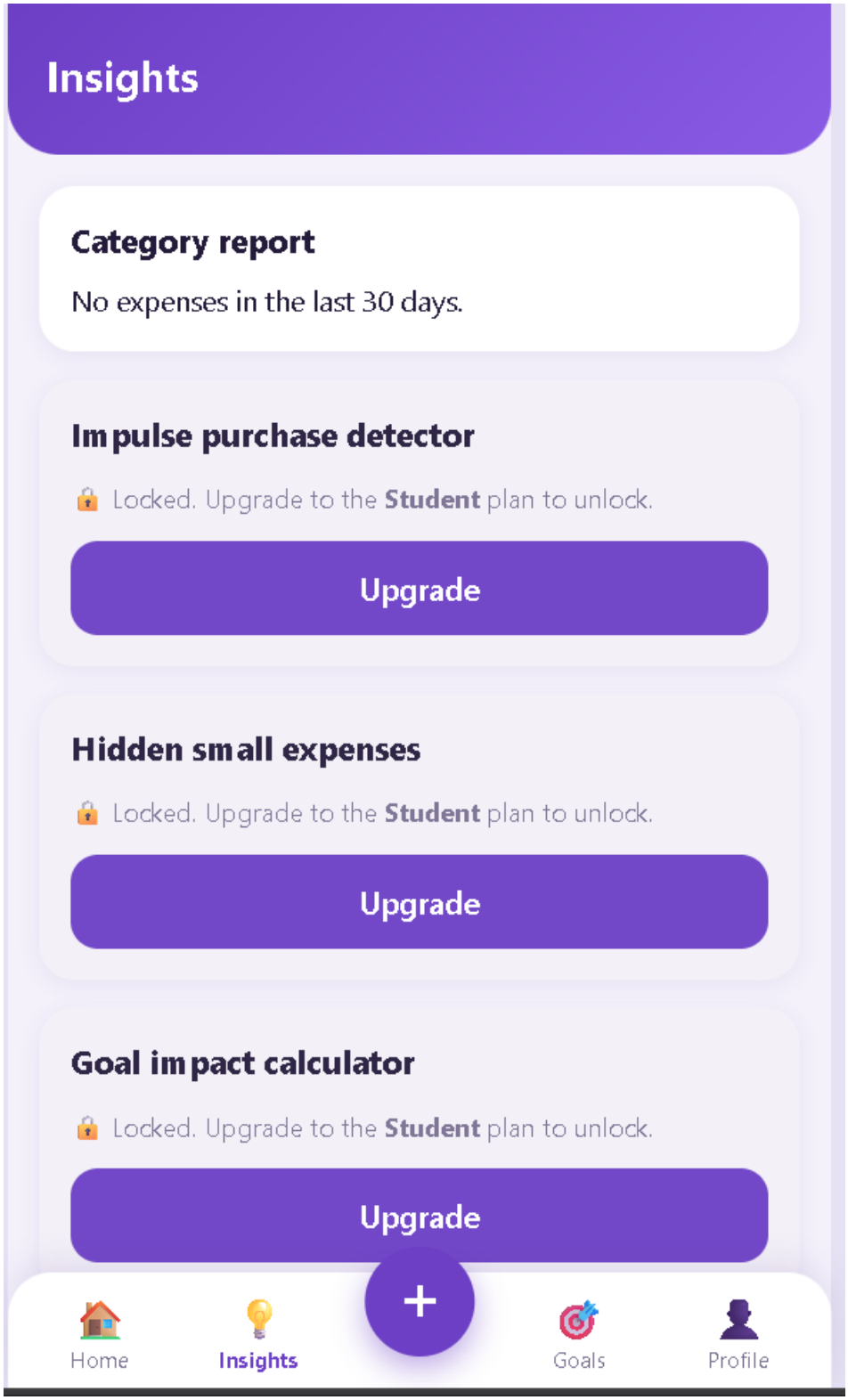
The Expense Tracker includes a documented JUnit 5 test suite for its core domain layer. Further verification of application workflows, database operations and integration is needed to establish the overall system results. Recording actual test outcomes provides evidence that the implemented requirements have been checked.

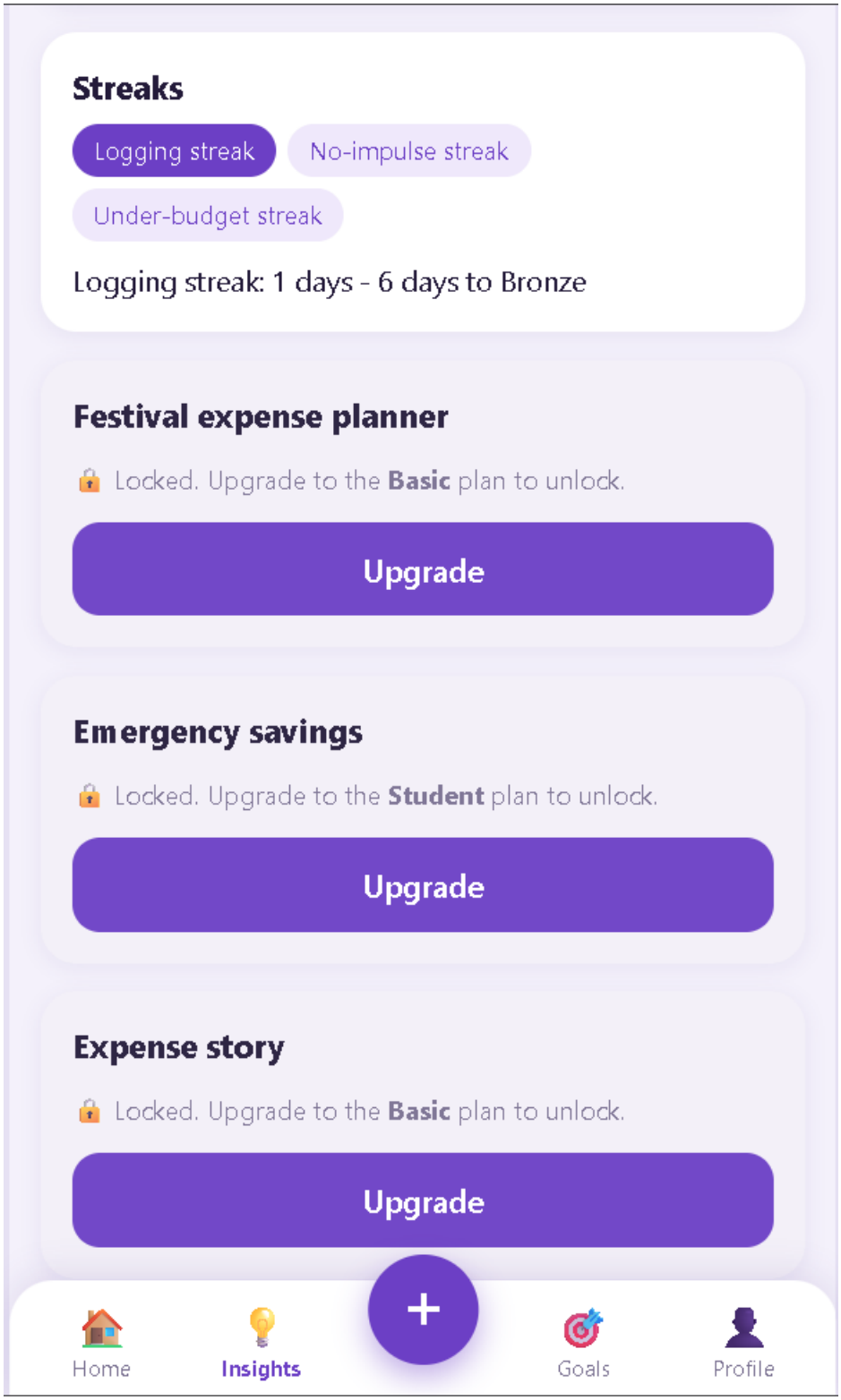
CHAPTER 10 – APPLICATION SCREENSHOTS AND USER MANUAL

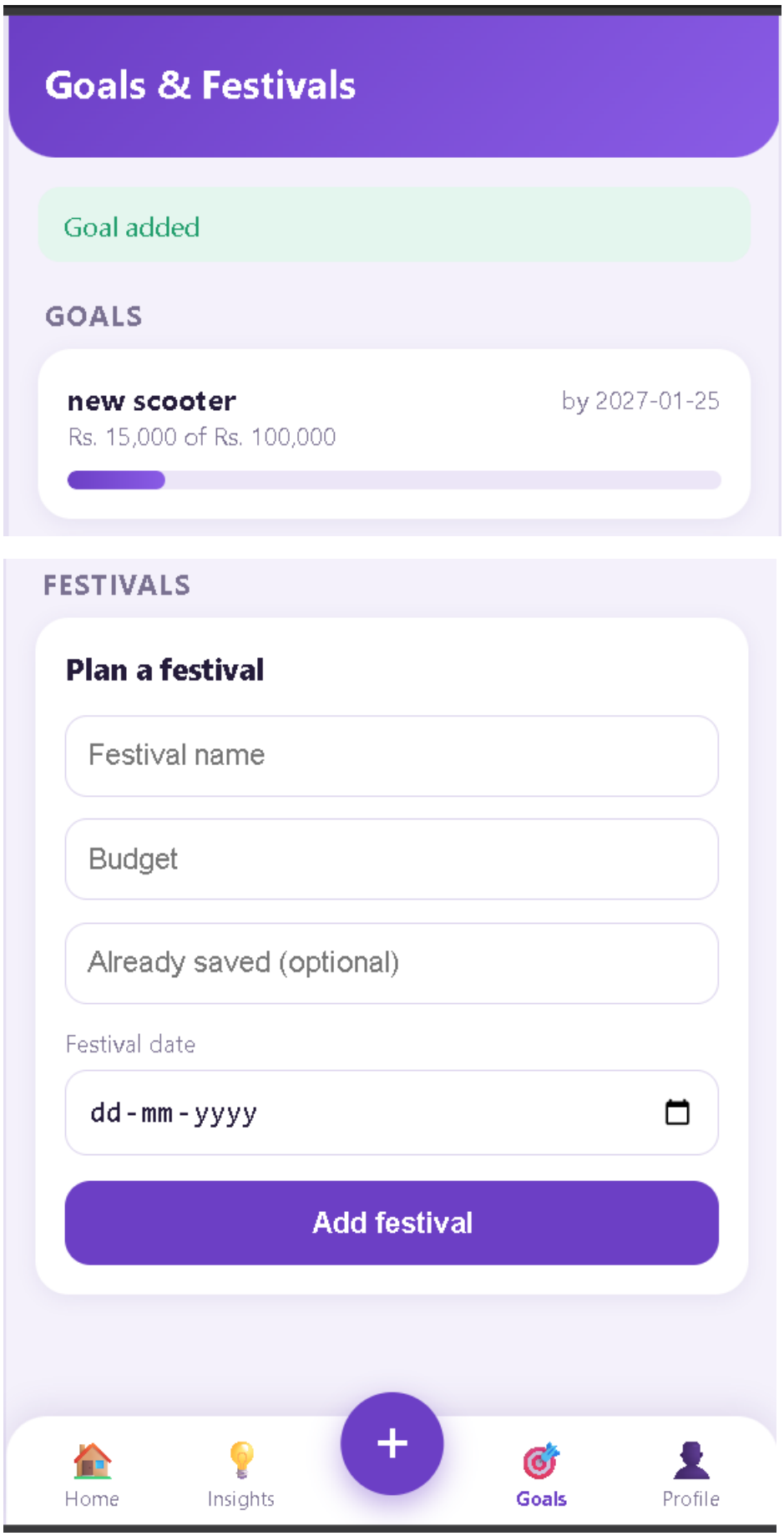



Hi, vinay sharma
Last 30 days
Free plan
Rs. 0
of Rs. 18,000 budget (0%)
Logging streak: 0 days - 7 days to Bronze
RECENT EXPENSES
No expenses yet. Tap + to add one, or load demo data.
Load demo data
Home
Insights
Goals
Profile


Hi, vinay sharma
Last 30 days
Free plan
Rs. 0
of Rs. 18,000 budget (0%)
Add expense
20000
EDUCATION
new phone
When (leave empty for now)
25-10-2026 20:16
Planned purchase
diwali
Save expense


Insights
Category report
No expenses in the last 30 days.
Impulse purchase detector
Locked. Upgrade to the Student plan to unlock.
Upgrade
Hidden small expenses
Locked. Upgrade to the Student plan to unlock.
Upgrade
Goal impact calculator
Locked. Upgrade to the Student plan to unlock.
Upgrade
Home
Insights
Goals
Profile


Streaks
Logging streak
No-impulse streak
Under-budget streak
Logging streak: 1 days - 6 days to Bronze
Festival expense planner
Locked. Upgrade to the Basic plan to unlock.
Upgrade
Emergency savings
Locked. Upgrade to the Student plan to unlock.
Upgrade
Expense story
Locked. Upgrade to the Basic plan to unlock.
Upgrade
Home
Insights
Goals
Profile


Goals & Festivals
Goal added
GOALS
new scooter
by 2027-01-25
Rs. 15,000 of Rs. 100,000
FESTIVALS
Plan a festival
Festival name
Budget
Already saved (optional)
Festival date
dd-mm-yyyy
Add festival
Home
Insights
Goals
Profile

Profile & Plans

vinay sharma
Current plan: Free

PLANS

Free **Rs. 0/mo**

30 expenses per month

✓ Category tracking
✗ Impulse purchase detector
✗ Hidden small expenses
✗ Goal impact calculator
✓ Streaks
✗ Festival expense planner
✗ Emergency savings
✗ Expense story

Student **Rs. 49/mo**

150 expenses per month

✓ Category tracking
✓ Impulse purchase detector
✓ Hidden small expenses
✓ Goal impact calculator
✓ Streaks
✗ Festival expense planner
✓ Emergency savings
✗ Expense story

Student ID (required)

Upgrade to Student

Basic **Rs. 149/mo**

Unlimited expenses

✓ Category tracking
✓ Impulse purchase detector
✓ Hidden small expenses
✓ Goal impact calculator
✓ Streaks
✓ Festival expense planner
✓ Emergency savings
✓ Expense story

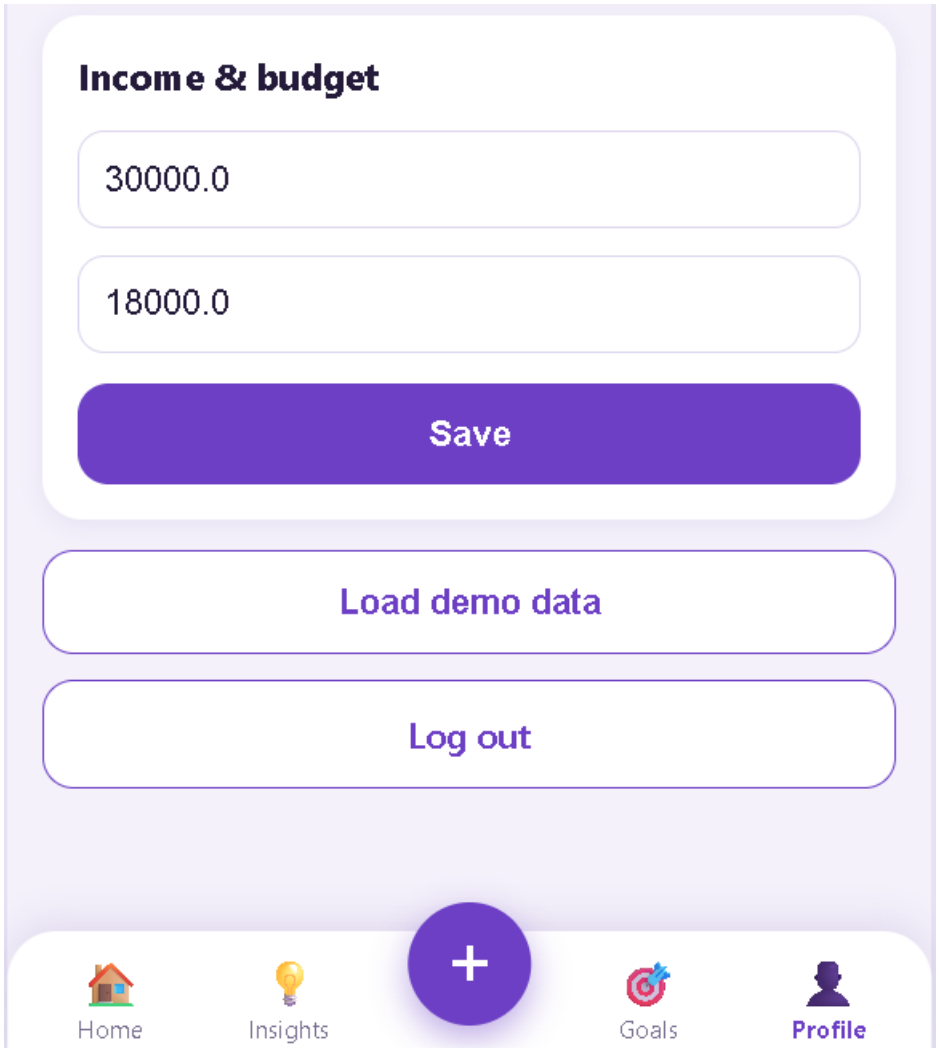
Upgrade to Basic

Emergency fund

Saved so far: Rs. 0

Deposit amount

Deposit


Income & budget
30000.0
18000.0
Save
Load demo data
Log out
Home
Insights
Goals
Profile

Insights

Category report

No expenses in the last 30 days.

Impulse purchase detector

No impulse purchases found. Nice discipline!

Hidden small expenses

No hidden small expenses found.

Goal impact calculator

new scooter (3 months left)
Needed per month: Rs. 28,333
Current monthly savings: Rs. 30,000
Shortfall: Rs. 0

CHAPTER 11 – CONCLUSION AND FUTURE ENHANCEMENTS

11.1 Conclusion

The Expense Tracker project focuses on improving personal expense management by helping users organise their expenses, understand spending patterns, and plan their finances. It aims to provide a simple and user-friendly approach to monitoring everyday spending and developing better financial awareness.

The project emphasises three main principles: behaviour-first spending awareness, goal-linked financial planning, and India-aware expense planning. Features such as expense categorisation, budget monitoring, savings goals, festival planning, and spending insights are intended to help users understand how their daily financial decisions affect their overall financial objectives.

The project also explores features such as the Impulse Purchase Detector, Hidden Small-Expense Finder, Narrative Expense Story, and subscription plans. These features aim to provide more meaningful insights than basic expense recording alone.

Overall, the Expense Tracker demonstrates how a personal finance application can support expense organisation, spending awareness, and savings planning. The final

outcomes depend on the features successfully implemented and tested during development.

11.2 Limitations

The project may have certain limitations depending on its implementation. Expense records and insights may depend on the accuracy and completeness of the information entered by users. Advanced analysis, automatic transaction tracking, and external financial integrations may require additional development.

Furthermore, proposed features that have not been implemented cannot yet provide their intended functionality. These limitations provide opportunities for future improvements.

11.3 Future Enhancements

The following enhancements may be considered for future development:

1. Automated Expense Tracking: Integrate supported services to reduce manual expense entry, subject to technical and privacy requirements.
2. Advanced Spending Analysis: Provide more detailed analysis of spending trends and recurring expenses.
3. Smart Notifications: Notify users about budget limits, unusual spending patterns, or savings milestones.

4. Improved Goal Planning: Offer additional tools for setting and monitoring multiple savings goals.
5. Enhanced Festival Planning: Provide custom budgets and expense plans for festivals and seasonal occasions.
6. Narrative Expense Insights: Generate easy-to-understand summaries based on recorded spending data.
7. Subscription Management: Implement different subscription plans with clearly defined features and access levels.
8. Data Export and Backup: Allow users to export expense reports and securely back up their financial records.
9. Enhanced Security: Introduce additional privacy and security measures to protect users' financial information.

These enhancements can be prioritised according to user requirements, technical feasibility, development time, and available resources.

11.4 Final Remarks

The Expense Tracker project provides a foundation for a personal finance application that goes beyond recording transactions by emphasising spending awareness and financial goal planning. With further development and testing, the proposed features could make the application more useful for students, young earners, and individuals seeking to manage their everyday expenses effectively.